

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.2: Logic Gates and Logic Circuits

Concept 3.2.1: Logic Gates (NOT, AND, OR, NAND, NOR, XOR)

Total Questions: 6

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w24_qp_13 (Q1.a)

- 1 (a) Describe the operation of a 2-input XOR gate.

.....
..... [1]

Question 2 | Source: 9618_s24_qp_12 (Q1.a)

- 1 (a) Describe the operation of each of the following logic gates:

NAND
.....
NOR
.....
XOR
.....
OR
..... [4]

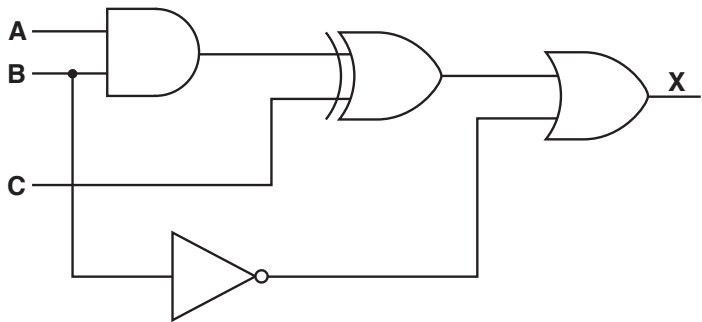
Question 3 | Source: 9618_s23_qp_11 (Q5.b)

- (b) Describe the function of the NAND and NOR logic gates.

NAND
.....
NOR
..... [2]

Question 4 | Source: 9618_w21_qp_11 (Q3.c)

- 3 A logic circuit is shown:



- (c) Identify **one** logic gate **not** used in the given logic circuit. Draw the symbol for the logic gate **and** complete its truth table.

Logic gate:

Symbol:

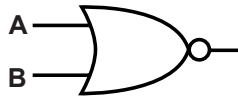
Truth table:

A	B	Output
0	0	
0	1	
1	0	
1	1	

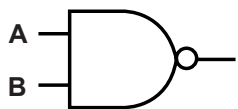
[3]

Question 5 | Source: 9618_w21_qp_12 (Q2.a)

- 2 (a) Complete the truth table for each of the following two logic gates.



A	B	Output
0	0	
0	1	
1	0	
1	1	



A	B	Output
0	0	
0	1	
1	0	
1	1	

[2]

Question 6 | Source: 9618_s21_qp_11 (Q8)

8 Tick (✓) **one** box in each row to identify the logic gate that each statement describes.

Statement	AND	NAND	NOR	XOR	OR
The output is 1 only when both inputs are 1					
The output is 1 only when both inputs are different					
The output is 1 only when both inputs are 0					

[3]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w24_ms_13 (Q1.a)

1(a)	1 mark for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	1
------	--	----------

Mark Scheme for Question 2 | Source: 9618_s24_ms_12 (Q1.a)

1(a)	1 mark for each correct answer: <i>NAND</i> • The output is 0 when both inputs are 1, otherwise the output is 1 <i>NOR</i> • The output is 1 when both inputs are 0, otherwise the output is 0 <i>XOR</i> • The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 <i>OR</i> • The output is 0 when both inputs are 0, otherwise the output is 1	4
------	--	----------

Mark Scheme for Question 3 | Source: 9618_s23_ms_11 (Q5.b)

5(b)	1 mark each <i>NAND</i> 0 is only output when both inputs are 1 // 1 is only output when none, or (either) one of the inputs is 1 <i>NOR</i> 1 is only output when both inputs are 0 // 0 is only output when (either) one or both inputs are 1	2
------	--	----------

Mark Scheme for Question 4 | Source: 9618_w21_ms_11 (Q3.c)

3(c)	1 mark for gate, 1 mark for matching symbol, 1 mark for matching truth table NOR	3
------	--	----------

Mark Scheme for Question 5 | Source: 9618_w21_ms_12 (Q2.a)

2(a)	1 mark for each completely correct truth table NOR <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>0</td></tr> <tr> <td>1</td><td>0</td><td>0</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table> NAND <table border="1"> <thead> <tr> <th>A</th><th>B</th><th>Output</th></tr> </thead> <tbody> <tr> <td>0</td><td>0</td><td>1</td></tr> <tr> <td>0</td><td>1</td><td>1</td></tr> <tr> <td>1</td><td>0</td><td>1</td></tr> <tr> <td>1</td><td>1</td><td>0</td></tr> </tbody> </table>	A	B	Output	0	0	1	0	1	0	1	0	0	1	1	0	A	B	Output	0	0	1	0	1	1	1	0	1	1	1	0	2
A	B	Output																														
0	0	1																														
0	1	0																														
1	0	0																														
1	1	0																														
A	B	Output																														
0	0	1																														
0	1	1																														
1	0	1																														
1	1	0																														

Mark Scheme for Question 6 | Source: 9618_s21_ms_11 (Q8)

8	1 mark per correct row	3																								
<table><tr><th>Statement</th><th>AND</th><th>NAND</th><th>NOR</th><th>XOR</th><th>OR</th></tr><tr><td>The output is 1 only when both inputs are 1</td><td>✓</td><td></td><td></td><td></td><td></td></tr><tr><td>The output is 1 only when both inputs are different</td><td></td><td></td><td></td><td>✓</td><td></td></tr><tr><td>The output is 1 only when both inputs are 0</td><td></td><td></td><td>✓</td><td></td><td></td></tr></table>			Statement	AND	NAND	NOR	XOR	OR	The output is 1 only when both inputs are 1	✓					The output is 1 only when both inputs are different				✓		The output is 1 only when both inputs are 0			✓		
Statement	AND	NAND	NOR	XOR	OR																					
The output is 1 only when both inputs are 1	✓																									
The output is 1 only when both inputs are different				✓																						
The output is 1 only when both inputs are 0			✓																							